

10 Academy: Artificial Intelligence Mastery

Week 11 Challenge Document

Forecasting Financial Inclusion in Ethiopia

Build a forecasting system that tracks Ethiopia's digital financial transformation using
time series methods

Date: 15 Jul 2026 - 21 Jul 2026

Overview

Business Need

You are a Data Scientist at **Selam Analytics**, a financial technology consulting firm specializing in emerging markets. Selam Analytics has been engaged by a consortium of stakeholders, including development finance institutions, mobile money operators, and the National Bank of Ethiopia, to develop a **financial inclusion forecasting system**.

Ethiopia is undergoing a rapid digital financial transformation. Telebirr has grown to over 54 million users since launching in 2021. M-Pesa entered the market in 2023 and now has over 10 million users. For the first time, interoperable P2P digital transfers have surpassed ATM cash withdrawals. Yet according to the 2024 Global Findex survey, only 49% of Ethiopian adults have a financial account; just 3 percentage points higher than in 2021.

The consortium wants to understand:

- What drives financial inclusion in Ethiopia?
- How do events like product launches, policy changes, and infrastructure investments affect inclusion outcomes?
- How did financial inclusion rates change in 2025, and how will it look like in the coming years, 2026 and 2027?

Your task is to build a forecasting system that predicts Ethiopia's progress on the two core dimensions of financial inclusion as defined by the World Bank's Global Findex:

1. **Access** — Account Ownership Rate
2. **Usage** — Digital Payment Adoption Rate

The Global Findex Framework

The Global Findex Database is the world's most comprehensive demand-side survey of financial inclusion, conducted every three years since 2011. Your models will forecast these Findex-defined indicators:

Access (Account Ownership)

"The share of adults (age 15+) who report having an account (by themselves or together with someone else) at a bank or another type of financial

institution or report personally using a mobile money service in the past 12 months."

Ethiopia's trajectory:

Year	Account Ownership	Change
2011	14%	—
2014	22%	+8pp
2017	35%	+13pp
2021	46%	+11pp
2024	49%	+3pp

Usage (Digital Payments)

"The share of adults who report using mobile money, a debit or credit card, or a mobile phone to make a payment from an account, or report using the internet to pay bills or to buy something online, in the past 12 months."

Ethiopia's indicators (2024):

- Mobile money account ownership: 9.45%
- Made or received digital payment: ~35%
- Used account to receive wages: ~15%

Your Deliverables

Build a system that:

1. **Understands and enriches** the provided financial inclusion dataset
2. **Analyzes** patterns and relationships in Ethiopia's inclusion data
3. **Models** how different type of national and regional level reports and events such as new policies, the launch of new products and services, etc. affect inclusion indicators
4. **Forecasts** Access and Usage for 2025-2027
5. **Presents** findings through an interactive dashboard

Data

Starter Dataset

You are provided with a structured dataset containing financial inclusion data for Ethiopia:

[ethiopia_fi_unified_data](#)

This dataset uses a **unified schema** where all records share the same structure. The [record_type](#) field indicates how to interpret each row:

record_type	Count	Description
observation	30	Measured values (Findex surveys, operator reports, infrastructure data)
event	10	Policies, product launches, market entries, milestones
impact_link	14	Modeled relationships between events and indicators
target	3	Official policy goals (e.g., NFIS-II targets)

Key design principle: Events are categorized by type (policy, product_launch, infrastructure, etc.) but are NOT pre-assigned to pillars. Their effects on specific indicators are captured through [impact_link](#) records. This keeps the data unbiased.

Supporting files:

- [reference_codes](#) — Valid values for all categorical fields
- [README.md](#) — Schema documentation

Supplementary Resource: Data Enrichment Guide

The following resource provides guidance on potential data sources and indicators for enrichment:

[Additional Data Points Guide](#)

This spreadsheet contains four sheets to help you identify useful data for your forecasting model:

Sheet	Description
A. Alternative Baselines	Additional data sources (IMF FAS, G20 indicators, GSMA, ITU, NBE, financial institution reports)
B. Direct Correlation	Indicators directly tied to inclusion: active accounts, agent density, POS terminals, QR merchants, transaction volumes, ATM density, bank branches
C. Indirect Correlation	Enabler/proxy variables: smartphone penetration, data affordability, gender gap, agent networks, urbanization, mobile internet, 4G coverage, literacy, electricity access, digital ID
D. Market Nuances	Ethiopia-specific context you should understand: P2P dominance (used for commerce, not just transfers), mobile money-only users are rare (~0.5%), bank accounts are easily accessible, very low credit penetration

Use this resource to:

- Identify additional observations to add in Task 1
- Understand which indicators matter most for forecasting
- Contextualize Ethiopia's unique market dynamics in your analysis

Learning Outcomes

Skills:

- Working with unified data schemas
- Exploratory data analysis with sparse time series
- Event impact estimation using comparable country evidence
- Regression modeling with intervention variables
- Forecasting with limited historical data
- Interactive dashboard development

Knowledge:

- Global Findex methodology and definitions
- Ethiopia's digital payment ecosystem (Telebirr, M-Pesa, EthSwitch, Fayda)
- Policy impact assessment approaches
- Working with uncertainty in data-limited contexts

Communication:

- Documenting data sources and methodology
- Presenting forecasts with appropriate confidence bounds
- Translating technical analysis into policy insights

Team

Tutors:

- Kerod
- Mahbubah
- Feven

Key Dates

- **Challenge Introduction** — 10:30 AM UTC on Wednesday, 15 Jul 2026
- **Interim Submission** — 8:00 PM UTC on Sunday, 19 Jul 2026
- **Final Submission** — 8:00 PM UTC on Tuesday, 21 Jul 2026

Communication & Support

- **Slack channel:** #all-week-11
- **Office hours:** Mon–Fri, 08:00–15:00 UTC

Tasks

Project Structure

```
ethiopia-fi-forecast/
├── .github/workflows/
│   └── unittests.yml
├── data/
│   ├── raw/                # Starter dataset
│   │   ├── ethiopia_fi_unified_data.csv
│   │   └── reference_codes.csv
│   └── processed/          # Analysis-ready data
├── notebooks/
│   └── README.md
├── src/
│   └── __init__.py
├── dashboard/
│   └── app.py
├── tests/
│   └── __init__.py
├── models/
├── reports/
│   └── figures/
├── requirements.txt
├── README.md
└── .gitignore
```

Task 1: Data Exploration and Enrichment

Objective: Understand the starter dataset and enrich it with additional data you find useful for the forecasting task.

Instructions:

1. **Understand the Schema**

- Load and explore [ethiopia_fi_unified_data.csv](#)
- Examine the structure: all records share the same columns
- **Sheet 1 (data):** Contains observations, events, and targets
 - [observation](#): Actual measured values from surveys, reports, operators
 - [event](#): Policies, product launches, market entries, milestones
 - [target](#): Official policy goals
- **Sheet 2 (impact_links):** Contains modeled relationships between events and indicators
- Review [reference_codes.csv](#) for valid field values

- Understand the challenges associated to setting **pillar** values to events (e.g., policy, product_launch)
 - Understand how **impact_link** records connect events to indicators via **parent_id**
2. **Explore the Data**
 - Count records by **record_type**, **pillar**, **source_type**, and **confidence**
 - Identify the temporal range of observations
 - List all unique indicators (**indicator_code**) and their coverage
 - Understand which events are cataloged and their dates
 - Review the existing **impact_links** and what relationships they capture
 3. **Enrich the Dataset** - Add new data that you believe will be useful for forecasting Access and Usage. Consider:
 - Additional observations: Are there other data points about Ethiopia's financial inclusion that could help? (e.g., disaggregations by gender or region from Findex microdata, additional years of infrastructure data, other relevant metrics)
 - Additional events: Are there policies, launches, or milestones that aren't captured but could affect inclusion? (e.g., regulatory changes, new partnerships, infrastructure investments)
 - Additional **impact_links**: Can you identify other relationships between events and indicators that should be modeled?
 4. **Follow the Schema**
 - For new observations: fill **pillar**, **indicator**, **indicator_code**, **value_numeric**, **observation_date**, **source_name**, **source_url**, **confidence**
 - For new events: fill **category** (e.g., policy, product_launch, infrastructure) but leave **pillar** empty
 - For new **impact_links**: fill **parent_id** (linking to the event), **pillar**, **related_indicator**, **impact_direction**, **impact_magnitude**, **lag_months**, **evidence_basis**
 5. **Document Your Additions**
 - For each new record, document:
 - **source_url**: Where you found the data
 - **original_text**: The exact quote or figure from the source
 - **confidence**: Your assessment (high/medium/low)
 - **collected_by**: Your name
 - **collection_date**: Date of collection
 - **notes**: Why you think this data is useful

Minimum Essential To Do:

- Merge necessary branches into main using a Pull Request (PR)
- Create branch "task-1"
- Commit work with descriptive commit messages
- Load all three datasets successfully
- Updated dataset with your additions and corrections
- [data_enrichment_log.md](#) documenting changes

Task 2: Exploratory Data Analysis

Objective: Analyze the data to understand patterns and factors influencing financial inclusion in Ethiopia.

Instructions:

1. Dataset Overview

- Summarize the dataset by [record_type](#), [pillar](#), and [source_type](#)
- Create a temporal coverage visualization: which years have data for which indicators?
- Assess data quality: distribution of [confidence](#) levels
- Identify gaps: which indicators have sparse coverage?

2. Access Analysis

- Plot Ethiopia's account ownership trajectory (2011-2024)
- Calculate and visualize growth rates between survey years
- If disaggregated data available:
 - Analyze the gender gap (male vs. female ownership)
 - Compare urban vs. rural ownership
- Investigate the 2021-2024 slowdown:
 - Account ownership grew only +3pp despite massive mobile money expansion
 - What factors might explain this deceleration?

3. Usage (Digital Payments) Analysis

- Analyze mobile money account penetration trend (2014-2024)
- Examine digital payment adoption patterns
- If data supports, explore:
 - The "registered vs. active" gap (registered accounts vs. survey-reported usage)
 - Payment use cases (P2P, merchant, bill pay, wages)

4. Infrastructure and Enablers

- Analyze available infrastructure data (4G coverage, mobile penetration, ATM density)
- Examine relationships between infrastructure and inclusion outcomes
- Identify potential leading indicators that might predict Findex outcomes

5. Event Timeline and Visual Analysis

- Create a timeline visualization showing all cataloged events
 - Overlay events on indicator trend charts
 - Visually identify apparent relationships:
 - Did account ownership accelerate after Telebirr launch (May 2021)?
 - Did mobile money accounts grow after M-Pesa entry (Aug 2023)?
 - What happened around Safaricom's market entry (Aug 2022)?
6. **Correlation Analysis**
- Examine correlations between different indicators
 - Which factors appear most strongly associated with Access?
 - Which factors appear most strongly associated with Usage?
 - What insights emerge from the existing [impact_link](#) records?
7. **Document Key Insights** Address questions such as:
- What factors appear to drive financial inclusion in Ethiopia?
 - Why might account ownership have stagnated (only +3pp) despite 65M+ mobile money accounts being opened?
 - What is the gender gap and how has it evolved?
 - What data gaps most limit your analysis?
 - What hypotheses emerge for testing in the impact modeling phase?

Note: Review Sheet D (Market Nuances) in the Additional Data Points Guide to understand Ethiopia-specific dynamics that may affect your interpretation.

Minimum Essential To Do:

- Merge branches from task-1 into main using PR
- Create branch "task-2"
- Commit work with descriptive commit messages
- EDA notebook with visualizations
- Summary of **at least 5 key insights with supporting evidence**
- Data quality assessment documenting limitations

Task 3: Event Impact Modeling

Objective: Model how events (policies, product launches, infrastructure investments) affect financial inclusion indicators.

Instructions:

1. Understand the Impact Data

- Load the [impact_links](#) sheet from your enriched dataset
- Join with events from the [data](#) sheet using [parent_id](#) to get event details
- Create a summary showing: which events affect which indicators, and by how much.

2. Build the Event-Indicator Matrix: Using the [impact_link](#) data:

- Each link specifies an event, the indicator it affects, the direction, magnitude, and lag

- Your task is to translate these relationships into a model that can predict how indicators change when events occur
 - Consider:
 - How do you represent an event's effect over time?
 - Do effects happen immediately or build gradually?
 - How do you combine effects from multiple events?
- 3. Review Comparable Country Evidence:** For events where Ethiopian pre/post data is insufficient, use documented impacts from similar contexts:
 - 4. Create the Association Matrix:** Build a matrix that summarizes:
 - Rows: Events
 - Columns: Key indicators (ACC_OWNERSHIP, ACC_MM_ACCOUNT, USG_DIGITAL_PAYMENT, etc.)
 - Values: The estimated effect of each event on each indicator
 - This matrix captures "which events affect which indicators and by how much."
 - 5. Test Your Model Against Historical Data:** Where possible, check if your impact model makes sense:
 - Telebirr launched in May 2021; mobile money accounts went from 4.7% (2021) to 9.45% (2024)
 - Does your model's estimated impact align with what actually happened?
 - If not, what might explain the difference?
 - 6. Refine Your Estimates**
 - Based on what you observe in the data, refine your impact estimates
 - Document your reasoning for any adjustments
 - Note which estimates you're confident about vs. uncertain about
 - 7. Document Your Methodology**
 - Explain how you modeled event impacts
 - What assumptions did you make?
 - What are the limitations of your approach?

Minimum Essential To Do:

- Merge branches from task-2 into main using PR
- Create branch "task-3"
- Commit work with descriptive commit messages
- Impact modeling notebook
- Event-indicator association matrix (table or heatmap)
- Documentation of:
 - Methodology and functional forms chosen
 - Sources for all impact estimates
 - Validation results comparing predicted vs. observed
 - Key assumptions and uncertainties

Task 4: Forecasting Access and Usage

Objective: Forecast Account Ownership (Access) and Digital Payment Usage for 2025-2027.

Instructions:

1. Define Targets

- **Account Ownership Rate** (Access): % of adults with account at financial institution or mobile money
- **Digital Payment Usage**: % of adults who made or received digital payment

2. Select Approach Given sparse data (5 Findex points over 13 years), consider:

- Trend regression (linear or log)
- Event-augmented model (trend + event effects)
- Scenario analysis

3. Generate Forecasts

- Baseline: Trend continuation
- With events: Incorporate expected developments
- Scenarios: Optimistic, base, pessimistic

4. Quantify Uncertainty

- Confidence intervals
- Scenario ranges
- Explicit acknowledgment of limitations

5. Interpret Results

- What does your model predict?
- What events have largest potential impact?
- What are the key uncertainties?

Minimum Essential To Do:

- Merge branches from task-3 into main using PR
- Create branch "task-4"
- Commit work with descriptive commit messages
- Forecasting notebook
- Forecast table with confidence intervals
- Scenario visualization
- Written interpretation

Task 5: Dashboard Development

Objective: Create an interactive dashboard that enables stakeholders to explore the data, understand event impacts, and view forecasts.

Instructions:

1. Dashboard Setup

- Use Streamlit (recommended) or Dash
- Create `dashboard/app.py`
- Include clear instructions in README for running locally

2. Dashboard Sections

- **Overview Page:**

- Key metrics summary cards (current values, trends)
- P2P/ATM Crossover Ratio indicator
- Growth rate highlights

- **Trends Page:**

- Interactive time series plots
- Date range selector
- Channel comparison view

- **Forecasts Page:**

- Forecast visualizations with confidence intervals
- Model selection option
- Key projected milestones

3. Technical Requirements

- At least 4 interactive visualizations
- Clear labels and explanations
- Data download functionality

4. Inclusion Projections Page:

- Financial inclusion rate projections
- Progress toward 60% target visualization
- Scenario selector (optimistic/base/pessimistic)
- Answers to consortium's key questions

Minimum Essential To Do:

- Merge branches from task-3 into main using PR
- Create branch "task-4"
- Commit work with descriptive commit messages
- Create working Streamlit application
- Include at least 4 interactive visualizations
- Display key metrics and forecasts
- Provide clear run instructions in README

Tutorials Schedule

In the following, the colour **purple** indicates morning sessions, and **blue** indicates afternoon sessions.

Wednesday (15 Jul 2026)

- Introduction to the Challenge (Kerod)
- Global Findex Framework (Feven)

Thursday (16 Jul 2026)

- Understanding the Unified Schema (Mahbubah)
- Building Association Matrices (Feven)

Friday (17 Jul 2026)

- Forecasting Approaches for Limited Data (Mahbubah)
- Q&A Session

Monday (20 Jul 2026)

- Q&A Session

Tuesday (21 Jul 2026)

- Q&A Session

Submissions

Interim Submission — Sunday (19 Jul 2026): 8:00 PM UTC

GitHub Repository containing: Completed Task 1 (enriched dataset with documentation) and Task 2 (EDA notebook with visualizations)

Interim Report including:

- Data enrichment summary (what you added and why)
- At least 5 key insights from EDA with supporting visualizations
- Preliminary observations on event-indicator relationships
- Data limitations identified

Feedback: You may not receive detailed comments on your interim submission, but you will receive a grade.

Final Submission — Tuesday (21 Jul 2026): 8:00 PM UTC

GitHub Repository containing: All tasks completed with clean, well-documented code

Final Report (blog post format, suitable for Medium):

- Executive summary
- Data and methodology description
- Key insights from exploratory analysis
- Event impact model methodology and results
- Forecasts for Access and Usage with uncertainty
- Dashboard description with screenshots
- Limitations and future work

Feedback: You will receive detailed comments and feedback in addition to a grade.

Other Considerations

- **Data Quality:** Document sources for any data you add
- **Uncertainty:** With limited historical data, wide uncertainty ranges are appropriate
- **Assumptions:** Make all assumptions explicit
- **Collaboration:** Use GitHub branches and pull requests effectively
- **Time Management:** Start early

References

Survey Data & Financial Inclusion Frameworks

- Global Findex Database — worldbank.org/globalfindex
- Global Findex Methodology — worldbank.org/globalfindex/methodology
- Global Findex Microdata — microdata.worldbank.org
- IMF Financial Access Survey — data.imf.org/FAS
- GSMA State of the Industry Report — gsma.com/sotir
- World Bank: Why Financial Inclusion Matters — worldbank.org/en/topic/financialinclusion/overview
- Demirgüç-Kunt et al. (2018): "The Global Findex Database 2017: Measuring Financial Inclusion and the Fintech Revolution" — openknowledge.worldbank.org
- IMF: Financial Inclusion and Economic Growth — imf.org/external/pubs/ft/sdn/2015/sdn1517.pdf

Ethiopia Sources

- National Bank of Ethiopia — nbe.gov.et
- EthSwitch S.C. — ethswitch.com
- Ethio Telecom — ethiotelecom.et
- Fayda Digital ID — id.gov.et
- Shega Media — shega.co

Predicting & Nowcasting Financial Inclusion (Proxy Approaches)

- World Bank: "Predicting Financial Inclusion with Machine Learning" — blogs.worldbank.org/opendata/predicting-financial-inclusion
- IMF Working Paper: "Financial Inclusion: Can It Meet Multiple Macroeconomic Goals?" — imf.org/en/Publications/WP
- CGAP: "Beyond the Findex: Tracking Financial Inclusion with Supply-Side Data" — cgap.org/blog/beyond-findex
- FSD Kenya: Using Administrative Data to Track Inclusion — fsdkenya.org/research
- Suri & Jack (2016): "The Long-Run Poverty and Gender Impacts of Mobile Money" (Kenya M-Pesa) — science.org/doi/10.1126/science.aah5309
- Blumenstock et al.: "Predicting Poverty and Wealth from Mobile Phone Metadata" — science.org/doi/10.1126/science.aac4420

Alternative Data for Financial Inclusion

- GSMA: Using Mobile Data for Financial Inclusion Insights — gsma.com/mobilefordevelopment
- World Bank: High-Frequency Monitoring of Financial Inclusion — worldbank.org/en/topic/financialinclusion/brief/high-frequency-monitoring
- IFC: Alternative Data and Financial Inclusion — ifc.org

Country Case Studies on Financial Inclusion Measurement

- Kenya: FSD Kenya FinAccess Surveys — fsdkenya.org/finaccess
- India: CRISIL Inclusix (Financial Inclusion Index) — crisil.com/en/home/our-analysis/reports/inclusix
- Rwanda: Access to Finance Rwanda — afr.rw
- Tanzania: FinScope Tanzania — fsdt.or.tz/finscope
- Pakistan: State Bank Financial Inclusion Index — sbp.org.pk

Association Matrices & Heatmaps

- Seaborn heatmap — seaborn.pydata.org/generated/seaborn.heatmap
- pandas crosstab — pandas.pydata.org/docs/reference/api/pandas.crosstab

Forecasting with Limited Data

- Forecasting: Principles and Practice (Ch. 5 & 7) — otexts.com/fpp3
- statsmodels OLS regression — statsmodels.org/stable/regression
- Confidence & prediction intervals — statsmodels.org/stable/generated/statsmodels.regression.linear_model.OLS.Results.get_prediction
- Matplotlib fill_between for uncertainty— matplotlib.org/stable/gallery/lines_bars_and_markers/fill_between

Dashboard Development

- Streamlit Documentation — docs.streamlit.io
- Plotly for Python — plotly.com/python